

Unit 3.5 Implicit Differentiation (p.208)

Note Title

4/9/2018

Use when you have to differentiate an unusual expression:
 $x^2 + xy + 3y^2 = 3$.

See y as $f(x)$: $x^2 + xf(x) + 3f(x)^2 = 3$.

Example: ① Find $\frac{dy}{dx}$ in the point $(0,1)$, where
 $x^2 + xy + 3y^2 = 3$.

$$\frac{d}{dx}(x^2 + xy + 3y^2) = \frac{d}{dx}(3)$$

$$\Rightarrow 2x + 1 \cdot y + x \frac{dy}{dx} + 6y \frac{dy}{dx} = 0$$

$$\Rightarrow (x + 6y) \frac{dy}{dx} = -2x - y$$

$$\Rightarrow \frac{dy}{dx} = \frac{-2x - y}{x + 6y}$$

$$x=0 \\ y=1$$

Find the tangent to $x^2 + xy + 3y^2 = 3$ at $(0,1)$.

Tangent: $y - y_0 = m(x - x_0)$
 $\Rightarrow y - 1 = m(x - 0)$

$$m = \left. \frac{dy}{dx} \right|_{x=0, y=1} = \frac{-2(0) - 1}{0 + 6(1)} = -\frac{1}{6}$$

$$\Rightarrow y - 1 = -\frac{1}{6}x \Rightarrow \boxed{y = -\frac{1}{6}x + 1}$$

2) $x \sin x + y \cos y = 2$. Find $\frac{dy}{dx}$

3) $x^3 + y^4 + x^5 y^7 = 8$. Find $\frac{dy}{dx}$.

4) Find $\frac{dy}{dx}$ if $\sin(2x) = \cos(3y)$

$$\frac{d}{dx} \sin(2x) = \frac{d}{dx} \cos(3y)$$

$$\Rightarrow \cos(2x) \times 2 = -\sin(3y) \times 3 \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{2 \cos(2x)}{3 \sin(3y)}$$

Find derivatives using implicit differentiation.

NB: $y = \arcsin x$ and $y = \arctan x$

(1) Find $\frac{dy}{dx}$ if $y = \arcsin x$.

Let $y = \arcsin(x)$

$$\Rightarrow x = \sin(y), y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

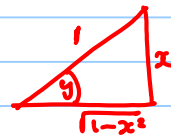
$$\frac{d}{dx}(x) = \frac{d}{dx}(\sin(y))$$

$$\Rightarrow 1 = \cos(y) \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\cos(y)}$$

$$= \frac{1}{\sqrt{1-x^2}}$$

$$\sin(y) = x.$$



$$\Rightarrow \cos(y) = \sqrt{1-x^2}$$

$$\Rightarrow \frac{d}{dx} \arcsin(x) = \frac{1}{\sqrt{1-x^2}}$$

(2) Find $\frac{dy}{dx}$ if $y = \arctan x$.

See textbook, p. 214.

No examples 3, 4, 5 in textbook.

Complete:

$$\frac{d}{dx} \sin^{-1}(f(x)) = \frac{d}{dx} \arcsin(f(x)) =$$

$$\frac{d}{dx} \tan^{-1}(f(x)) = \frac{d}{dx} \arctan(f(x)) =$$

Examples:

$$1) y = \sin x^2. \quad \frac{dy}{dx} = \cos(x^2) \times 2x$$

$$2) y = \arcsin x^2. \quad \frac{dy}{dx} = \frac{1}{\sqrt{1-x^4}} \times 2x = \frac{2x}{\sqrt{1-x^4}}$$

$$3) y = e^{\arctan x} \quad \frac{dy}{dx} = e^{\arctan(x)} \times \frac{d}{dx}(\arctan(x)) \\ = e^{\arctan(x)} \times \frac{1}{1+x^2}$$

$$4) y = \arctan e^x. \quad \frac{dy}{dx} = \frac{e^x}{1+e^{2x}}$$

Text book: Example 3, p. 212.
Example 5, p. 214.

